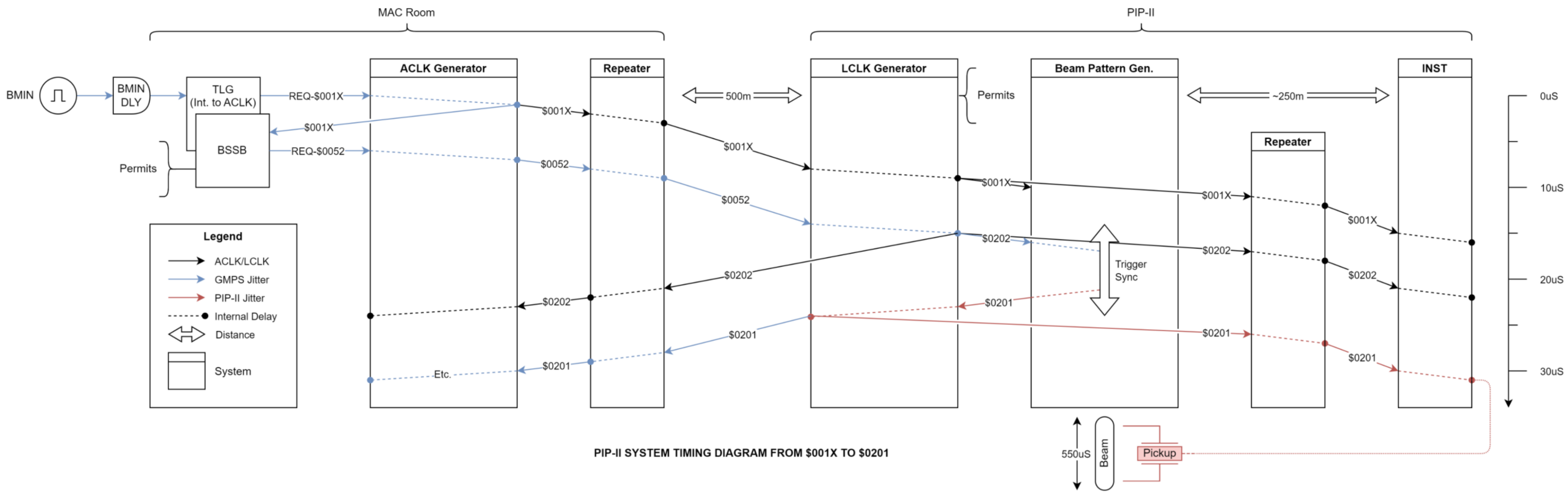
What the heck is Evan talking about.

A quick overview for Instrumentation of:
ACORN-doc-2204-v3: Proposal for Precision Machine Timing (fnal.gov)

September 19, 2024

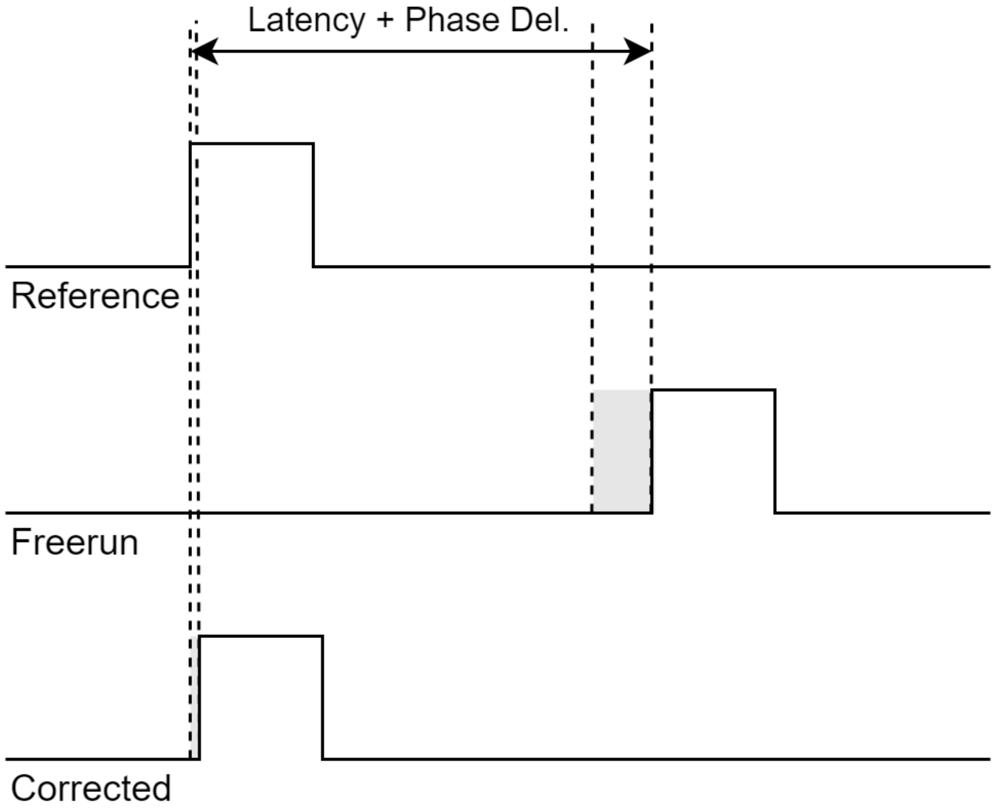
Background:

- Inter-bunch timing is difficult
- Distributed inter-bunch timing is even more difficult
- Addition of a global reference signal will improve reliability



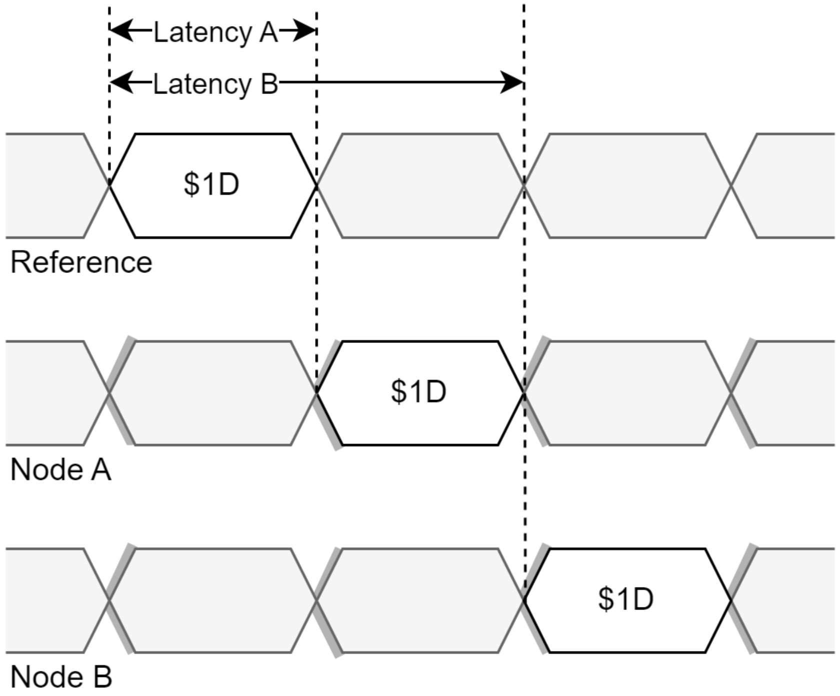
Concept 1: Global Reference (PPS)

- Pulse Per Second (PPS, pronounced peepis)
- Plays out at every node simultaneously
- Orients nodes relative to the grandmaster (ACLK generator)
- Tracks drift, changes in network conditions
- Timestamp directly to FPGA



Concept 2: Event Windows (EW)

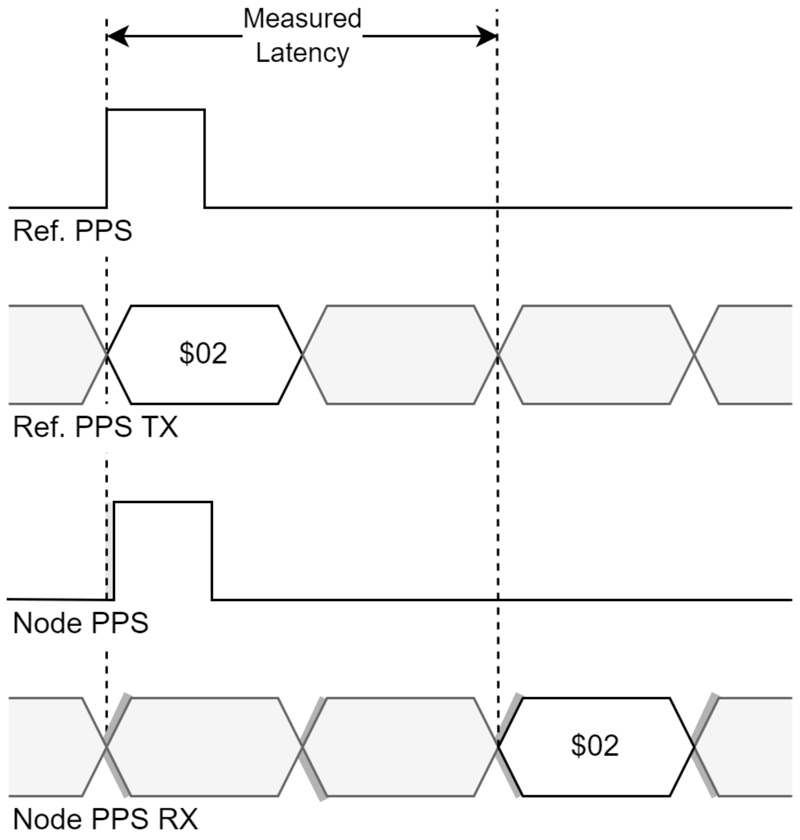
- Window during which event may play out (time is discrete)
- Window can be tied to RF (9.027MHz) or global 10MHz.
- *Should* work for other Beam Synchronous clocks (MI/Boo)
- Compensates for network jitter, windows >> network drift.



Discrete Window Alignment

Please take these concepts at face value.

- Evan is working very hard to make sure they are implemented reliably
- Provides rigid, deterministic timelines with integer offsets
- Current LCLK protocol still stands!
- Addition of these clocks ensures it is accurate to the ns.

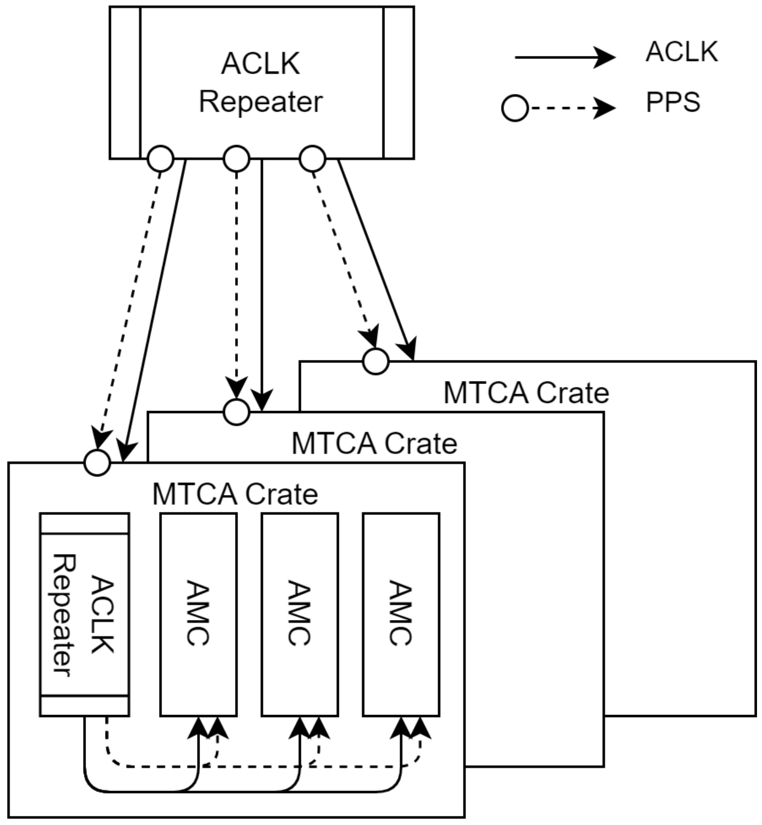


Event Window Offset Measurement

MTCA Integration

- You do not need a White Rabbit receiver!
- Pulse per second/ EW clocks can be provided by an MFTU
- Clocks can be fanned out through the backplane
- Events can be fanned out through the backplane

  (16b events over 100Mb LVDS)

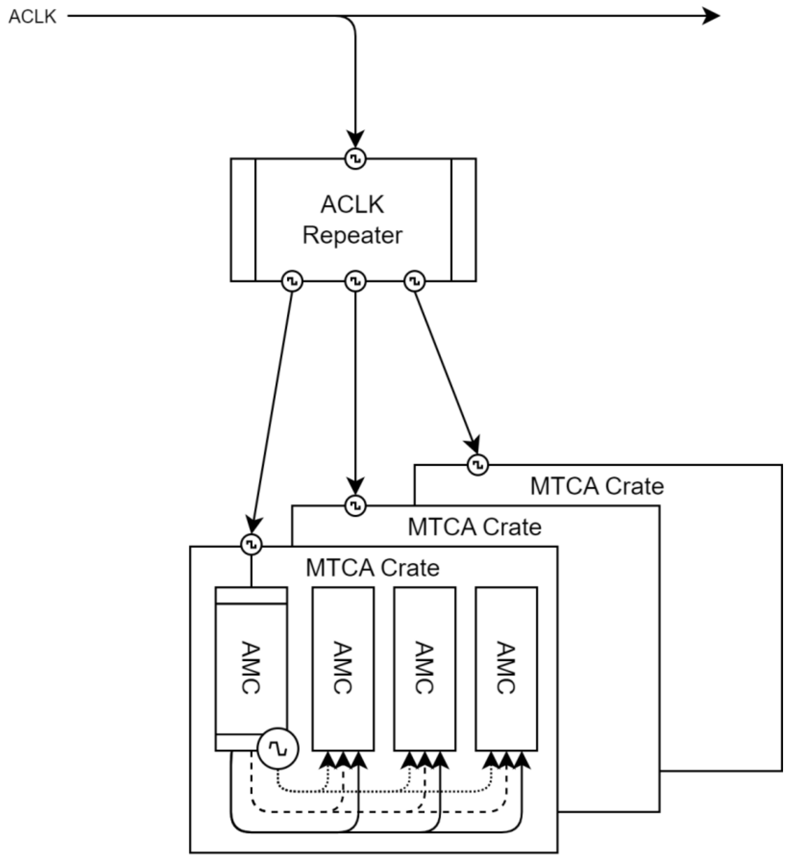


Integration Without WR Hardware

Synchronization through ACLK/LCLK

ACLK Integration (No PPS or EW)		
Parameter	Value	Notes
Network Latency	5us	(Propagation Delay)
Network Jitter	~ 10ns	(Repeater Config.)
Decoder Jitter	10ns	(SERDES)
Offset Error	> 5us	
Event Jitter	> 20ns	

- PPS *can* be inferred from sync event (5uS ±20ns error)
- 10MHz EW *can* be generated locally (±30ns error)
- Okay for most applications, comparable to TCLK

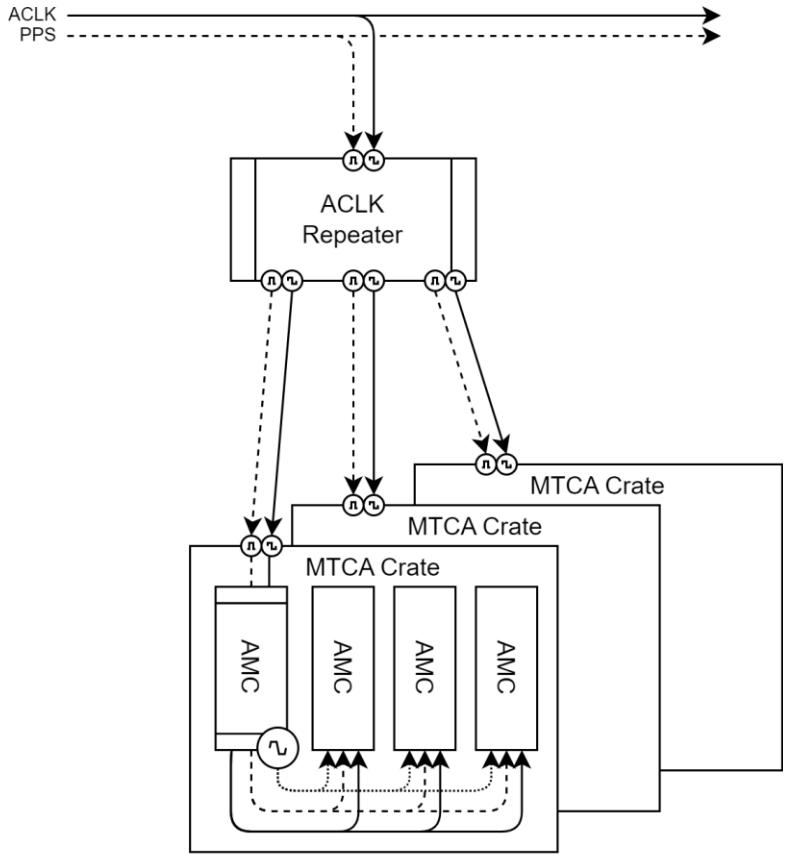


Local EW / PPS Generation

Synchronization through ACLK with PPS

ACLK Integration With PPS		
Parameter	**Value**	**Notes**
Network Latency	5us	
Network Jitter	~ 25ns	
Decoder Jitter	10ns	
Offset Error	< 1ns	(Corrected for with WR)
Event Jitter	~10ns	(10MHz EW Generation)

- Network latencies are corrected via PPS
- 10MHz EW *can* be generated locally against PPS
- Provides resilient time-domain synchronization

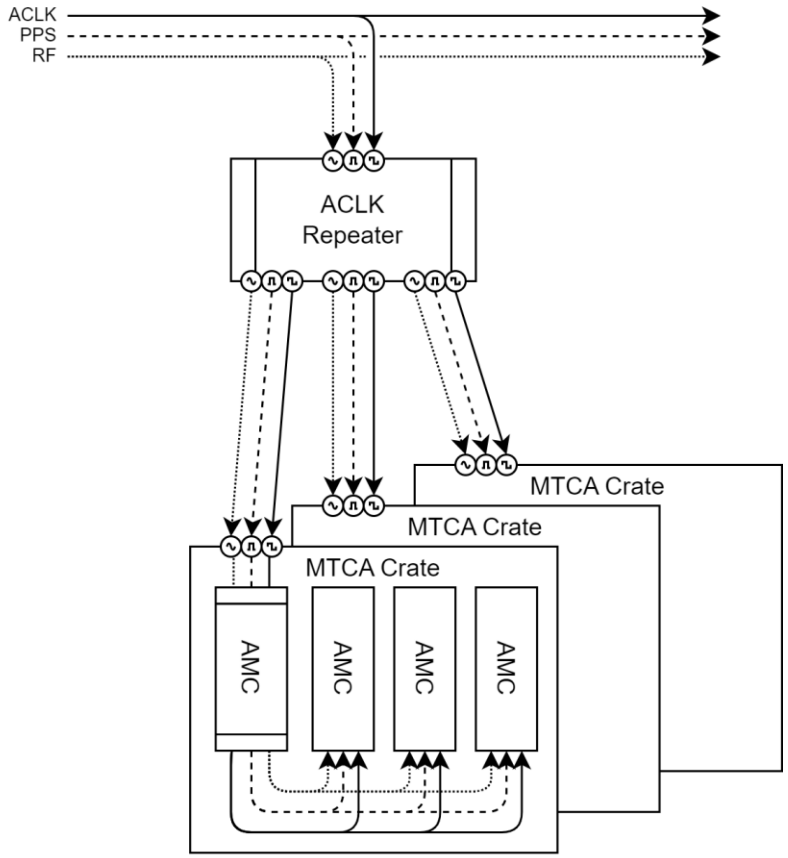


PPS & Local EW Generation

Synchronization through ACLK with PPS & RF

ACLK Integration With PPS and EW		
Parameter	**Value**	**Notes**
Network Latency	5us	
Network Jitter	~ 25ns	
Decoder Jitter	10ns	
Offset Error	< 1ns	(Corrected for with WR)
Event Jitter	< 1ns	(Sourced from RF Ref.)

- Network latencies are corrected via PPS
- RF synchronization is maintained via Event Windows
- Provides resilient RF-domain synchronization.



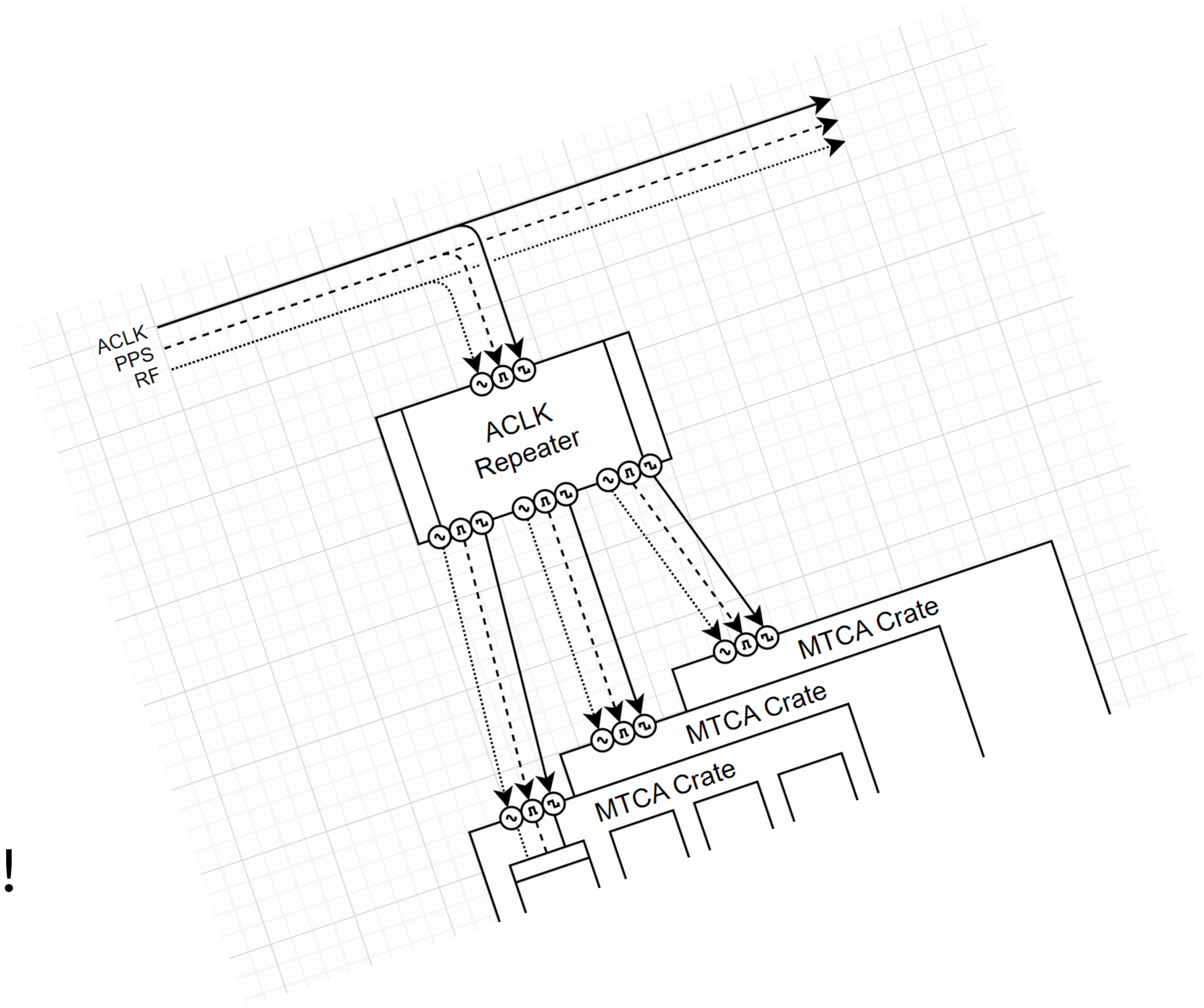
PPS & RF Integration

But now there are 3 signals going to my crate.

I am sorry.

General Rationale:

- More infrastructure, *but* very robust implementation
- Instrumentation can start integration immediately
- PPS/ACLK is good enough for most applications.
- RF event windows should work for Booster synchronous clocks!



How much will PPS cost Controls?

- 32ch fanouts are $13K
- Standalone nodes are $2K
- Standard fiber interconnects
- WR hardware required at:
  ACLK Generator
  LCLK Generator
  Laser Wire
- WR hardware every 50m would be a nice to have.

Thank you.

-Evan